SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

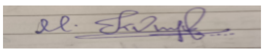
Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I *Kumaran M(192525022)* certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Scenario-Based Questions CO-AT2

QUESTIONS: 05 **Total Marks:25**

1. **Unsupervised Training of Neural Networks, Auto Encoders**
   A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. **Auto Encoders, Representation Learning**
   A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. **Width and Depth of Neural Networks, Activation Functions**
   A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. **Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks**
   A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. **Deep Learning Applications, Representation Learning**

   A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are wellstructured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Answers:

1) **Scenario:**

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Solution:

I recommend using an Autoencoder, an unsupervised deep learning model designed to learn the normal patterns present in unlabeled data. Since the hospital has thousands of ECG records without labels, an autoencoder is an ideal solution because it does not require manually classified training data.

The autoencoder consists of two main components: an encoder, which compresses the input ECG signal into a lower-dimensional representation, and a decoder, which reconstructs the original ECG signal from the compressed features. During training, the model learns to accurately reconstruct normal ECG patterns. When an abnormal ECG is given as input, the reconstruction error becomes significantly higher because the model has not learned those abnormal patterns. This reconstruction error can be used to identify anomalies. **Workflow**

1. Collect unlabeled ECG records.
2. Preprocess the ECG signals by removing noise and normalizing the data.
3. Train the autoencoder using the ECG dataset.
4. Encode the ECG into compressed features.
5. Decode the compressed representation to reconstruct the ECG.
6. Calculate the reconstruction error.
7. If the reconstruction error exceeds a predefined threshold, classify the ECG as abnormal; otherwise, classify it as normal.

Advantages

- Works with unlabeled data.
- Detects rare and unknown abnormalities.
- Reduces manual labeling effort.

- Provides accurate anomaly detection.

2) Scenario:

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Solution:

An **Autoencoder** can efficiently reduce the dimensionality of customer purchase data by learning a compact representation of the original features. High-dimensional purchase data often contains redundant or correlated information, making customer segmentation less effective.

The encoder compresses the original customer purchase information into a lower-dimensional feature vector while preserving the most important purchasing patterns. These compressed features are then used as input for clustering algorithms such as K-Means to group customers with similar buying behavior.

Working Process

1. Collect customer purchase history.
2. Normalize the data.
3. Train the autoencoder using customer purchase records.
4. Extract compressed features from the encoder.
5. Use the compressed features for customer segmentation.
6. Apply clustering algorithms to identify customer groups.
7. Use the identified groups for targeted marketing and personalized recommendations.

Advantages

- Removes redundant features.
- Improves clustering accuracy.
- Reduces computational complexity.
- Learns meaningful customer representations automatically.

3) **Scenario:**

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Solution:

A shallow neural network has limited capability to learn complex facial features. To improve performance, the network architecture should be modified by increasing its depth and selecting appropriate activation functions.

The system should use multiple hidden layers so that the network can learn low-level features such as edges and textures in the initial layers and more complex facial structures in deeper layers. The **ReLU (Rectified Linear Unit)** activation function should be used in hidden layers because it speeds up training and helps overcome the vanishing gradient problem. The **Softmax** activation function should be used in the output layer for multi-class face classification.

Additional improvements include Batch Normalization to stabilize training and Dropout layers to reduce overfitting.

Recommended Changes

- Increase the number of hidden layers.
- Increase the number of neurons where necessary.
- Use ReLU activation in hidden layers.
- Use Softmax activation in the output layer.
- Apply Batch Normalization.
- Use Dropout for regularization.

Benefits

- Learns complex facial features. • Improves recognition accuracy.
- Reduces overfitting.
- Faster and more stable training.

4) **Scenario:**

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Solution:

A **Restricted Boltzmann Machine (RBM)** is an unsupervised neural network used for collaborative filtering and recommendation systems. It learns hidden relationships between users and movies from viewing history without requiring labeled data.

The streaming service first creates a user–movie interaction matrix representing watched or rated movies. The RBM learns hidden features that represent user preferences and movie characteristics. Based on these learned relationships, the RBM predicts which movies a user is likely to enjoy and recommends those movies.

Workflow

1. Collect user viewing history.

2. Create a user–movie interaction matrix.
3. Train the RBM using viewing data.
4. Learn hidden user preferences and movie features.
5. Predict unseen movie ratings.
6. Recommend movies with the highest predicted preferences.

Advantages

- Works with unlabeled data.
- Learns hidden user interests.
- Handles sparse datasets effectively.
- Generates personalized movie recommendations.

5) **Scenario:**

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Solution:

A deep learning solution can be designed using representation learning and a Deep Neural Network (DNN) or an Autoencoder. Representation learning automatically extracts meaningful features from raw sensor readings without requiring manual feature engineering.

The collected sensor data is first cleaned and normalized. An autoencoder learns compressed representations of normal sensor behavior. These learned features are then supplied to a deep neural network classifier that identifies whether a product is defective or non-defective. Whenever abnormal sensor patterns are detected, the system generates an alert for quality inspection.

System Design

1. Collect sensor readings continuously.
2. Preprocess and normalize the data.
3. Train an autoencoder to learn useful feature representations.
4. Extract compressed features from the encoder.
5. Train a deep neural network classifier using the learned features.
6. Detect defective products in real time.
7. Generate alerts and store inspection reports.

Advantages

- Automatic feature extraction.

- Real-time defect detection.
- Improved manufacturing quality.
- Reduced manual inspection costs.
- Increased production efficiency.